Pal et al. 2021 — Breast Cancer

10x · 45 samples · 305,224 cells

CCCA cell-state manifold · malignant embryonic-program analysis

59,356 malignant cells analyzed (22 samples)

GO / NO-GO GATES

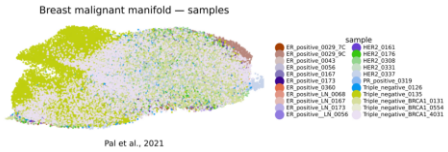
Gate 1 **GO** Mixing + continuity judged on the regenerated embedding. iLISI 1.0 → 4.96 of 22 samples (Harmony θ=10); one continuous connected component (11 speck cells removed). A proliferation compartment (76% cycling, 14,165 cells) is resolved before the GO → the trajectory is scoped to the low-cycling developmental Body (45,191 cells).

Gate 2 **GO** Module geography — the non-branching case. No Gast/pSom two-arm hub; a single embryonic axis runs pSom-high (origin) → Crest-high (terminus). Smoothed pSom vs Crest r = −0.93; joint pSom&Crest enrichment 0.23 (<1, inverse gradient). Crest terminal (high-decile pseudotime 0.92); Gast weak and flat.

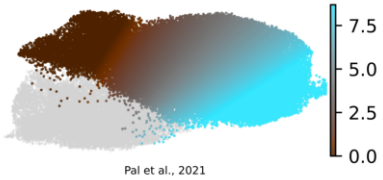
Gate 3 **NA** Branching-only gate → not applicable (single linear axis, no bifurcation to test for cross-repression).

STEP FIGURES

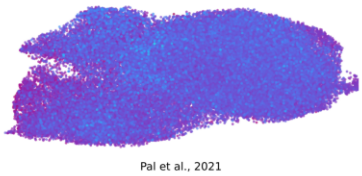
01 · Sample integration



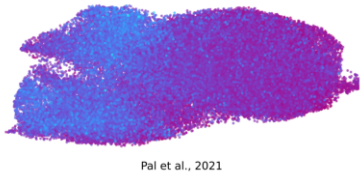
02 · Pseudotime manifold



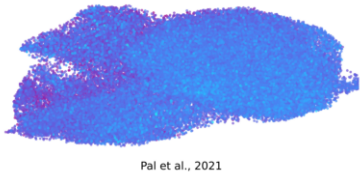
03 · Gastrulation



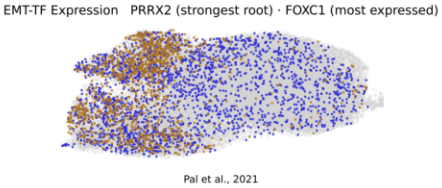
03 · Somitogenesis (origin)



03 · Neural crest (terminus)



04 · EMT-TF (PRRX2 / FOXC1)



PARAMETER CHOICES & RATIONALE

Min-cell threshold: ≥12 — geometric mean of the 56→3 spillover cliff (ratio 18.67). Drops 4 samples contributing 1–3 malignant cells; keeps 22 of 26.

Integration: Harmony on sample, θ=10 (swept 2/5/10; same-sample neighbour frac 0.51→0.43→0.43, sharp then flat). n_pcs=20, spread=1.3, overdispersed genes=3733.

Sub-region scoping: Leiden res=0.15 → two lobes. Cycling lobe (76% cycling, prolif +0.36) excluded; whole-manifold root would land 77%-cycling (ρ vs proliferation = −0.52).

Geometry: Non-branching curve (scFates, N=20, epg_λ=0.3): 2 tips, 0 forks, fold-free (self-approach 2.98). Fit on the Body only, on the joint embedding (no re-embedding).

Root node: Curve tip 0 — pSom/Gast-high, Crest-low, 9% cycling. Pseudotime not proliferation-driven (ρ=+0.01). Crest is developmentally late, so it never roots.

EMT-TFs: PRRX2 (strongest root) & FOXC1 (most expressed, 16%) from a 10-TF panel gated on detection — all detected EMT-TFs are origin-enriched (no terminus factor). VIM excluded.

Module strength: Gast μ = −0.007 · pSom μ = +0.036 · Crest μ = −0.034 → Linear-pSom/Crest (no hub; Gate-3 hub r not applicable).

VERDICT The embryonic pattern reproduces as the NON-BRANCHING case — one continuous, sample-integrated malignant manifold carrying a single somitogenesis→neural-crest developmental axis, with no Gast/pSom bifurcation. Gates 1 & 2 GO, Gate 3 NA.